Speech Recognition

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Agenda

Work Done at Flipkart-Walmart:

- Context, Problem Statement, Requirements
- Research, Architecture Selection
- Data, Feature, Modeling
- Training, Offline/Online Evaluation
- Error Analysis, Online Testing, Deployment
- Post Deployment, Operationalise
- Challenges
- Next Steps, References

Today

- Transformer based approaches

Context, Problem Statement, Requirements

Context: Flipkart (Walmart invested 18b$ in 2018). Leading eCommerce company in India with millions of Users
My Role: Tech Lead: supporting 4 applied research data scientist (50% research + development + 50% supporting team and cross functional collaboration (with Chatbot team, UXR, Business, Product, Engineering, Data Labeling, Operations teams)

Problem / Requirement:

- User Growth - additional 200+ million customer who were using smartphones, but didn't know how to use App in English
- Provide Voice Assisted Shopping in Hindi, and other Indian languages

Timeline:

- GTM: Had to launch before Festive season in Oct
- Develop in six months, testing for two months

Success criteria:

- Successful order placed using voice

Functional Requirement:

- V1 support Hindi + English (Delivery Goal)
    - Metrics: Transaction success, WER < 5%
- V2 Approach for 10 languages (Discovery Goal)
    - Kannada, Telugu,

Non-Functional Requirements:

- Performance: Latency - e2e <300ms
- Scale: Able to handle peak load
- Logging, Retraining
- Monitoring, Alerting, Escalation

Research, Architecture Selection

Research:

RQ: Which approaches are suitable for short-mid term

- Hindi, English, Code-mixed
- Low resources languages

Selection Criteria:

- Deployed Solution at Scale
- Ease of Implementation
  - Code available
- Long term continuity based on pioneering work
  - Google, Meta, MILA,

Researched + Selection

- Researched 200+ approaches to get deeper understanding just beyond SOTA approaches
  - Which can help in come up with different idea/ Technique
- Selected top-2 approaches in 5 categories based on arch
  - Transformer, CNN, LSTM, HMM-DNN, Hybrid
- Presented to wider Applied Research Science team (60+ people) in Brown bag session and got inputs
- Collaboration opportunity with Transcription, Translation and TTS teams

Selected Approaches:

- Transformer based, LSTM based
- HMM-DNN (RWTH Aachen University)

Data Strategy, Feature Engineering, Modeling (1/2)

Begin with English dataset:

- 960hrs LibriSpeech Clean and Noisy data

Generate Labeled data:

- Recording in Studio
- Record in Real world with noise
- TTS from Google, Inhouse TTS

Reduce data:

Based on ROI of further training and WER impact,

- Remove Sentences, Words and Phrases
- Eliminated upto 40% of data

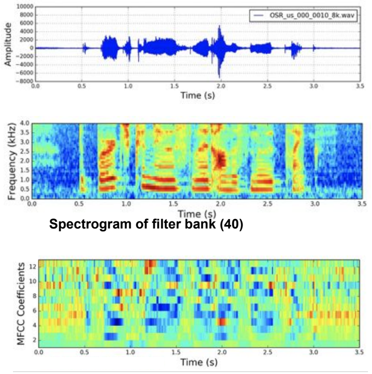
Data Augmentation + Robust Training

Each variation helps in doubling the data

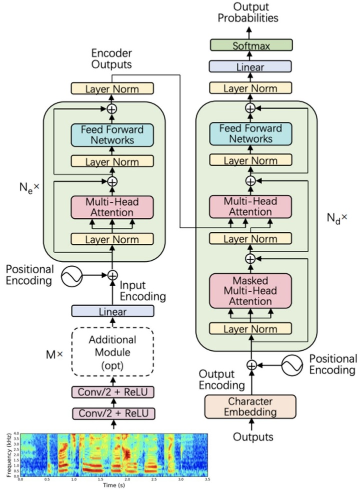
- Varying Speed, Tempo, Gain
- Vocal Tract Length Perturbation
- Adding environmental noise (door bell, honk)
- Frequency, Time masking and Time warp

Golden dataset:

- Create and update for testing across various versions of ASR system



Feature Engineering, Modeling (2/2)



Parameters:

- No. of layers:
  - 16, 20, 24, 32, 36, 48
- Embedding dimension:
  - 512, 768, 1024, 1536, 2048
- No. of heads:
  - 4, 8

Subword:

- **Phonemes 73**, Subword 5k/10k

Language Model:

- 3-gram, 4-gram
- Gated CNN based

Inference:

- Beam size

Audio signal:

20/25ms (10ms stride)

30ms, 50ms, 32ms, 100ms (5ms shift)

Filters:

40, 80, 120 Filter bank

Gammatone filters

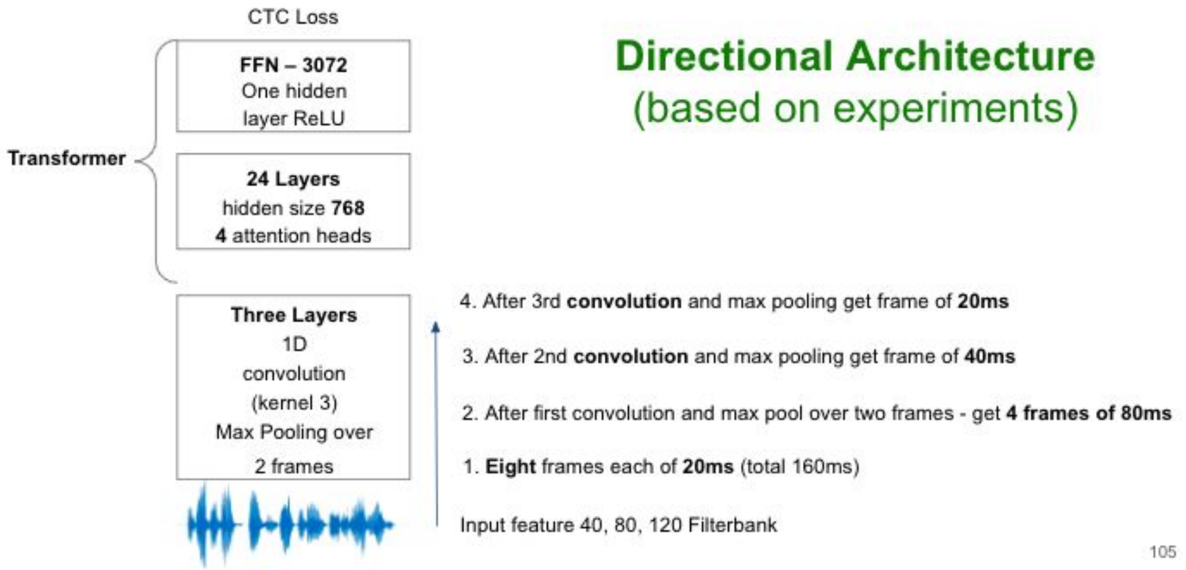
Feature Extraction:

Conv-1D, MaxPooling, VGGNet

Features from Raw waveform: SincNet

Offline test and Directional Architecture (1/2)

Transformer based Models (Paper & 100 iterations numbers)	test-clean WER	test-other WER	Epochs	Comment
CTC Transformer (ngram-LM) - paper	2.92	6.65	320	Could take base arch from paper and compare
CTC Transformer (ngram-LM) - our run	4.41	11.39	100	
CTC Transformer (gcnn-LM) - paper	2.55	5.91	320	With limited 100 iterations, cnn-LM gives improvement
CTC Transformer (gcnn-LM) - our run	3.75	9.92	100	



Error Analysis, Online testing, Deployment

Error Analysis:
At both Word and phoneme level
- Wrong Pronunciation
- Repeated words
  - Slipper, *Chappal (in Hindi)*
- Intent -Long phrase to express intent
  - Hair dryer *(in Hindi said long sentence - bal sukhane value machine)*

Noisy:
- Background noise
- Multiple speakers
- Overlapping Speaker
- Very noisy environment

Iterate Experiment::
Analyse and classify error for improving the model

Tools:
Analysing results (get WER, Successful transaction rate)
System metrics (latency)
Logging:
- Log data and predictions
- Analyse at detailed level

Monitoring and Alert - for troubleshooting

Testing:
Eat your own food (internal)
Alpha and beta (iterate with small user segment)
- Roll back switch if need to revert back

Deployment Rollout:
Cold start of new items added on daily basis
- Create hard coded dictionary initially
- Later include audio for new items

Post Deployment, Operationalise

Ramp up deployment:
- Initially hourly manual monitoring

Monitoring and Alerting
- Broken pipeline to log data
- High WER
- Lower Successful Transactions

Daily Report on different components

Operationalise whole system with team responsibility
- Chat bot, ASR, Engineering, Infra

Handbook for Troubleshooting:
- Using Monitoring Dashboard to understand what went wrong
- How to deep dive
- Whom to escalate

Challenges

Errors because of Background Noise, Multispeaker
- COVID - had more impact as background voice by (TV or other family members)
- Augmented data with these and further trained

Repeated words, Code- Mixed and Intent in long sentence
- Language identification, translation and semantic understanding

Not all required information present in the voice
- Had to iterate to find out missing entity and prompt user again (chat bot enhancement)]
- Give me 2kg milk (don’t specify brand or type of milk)

New words
- Use LM or Hard coded dictionary to map Acoustic model output

Next Steps

References - Techniques (2021)

Additional requirement surfaced as immediate priority

- **Streaming**
  - Two pass
  - Shared and independent
- Code-switch
- Multi-Speaker
- Background Noise
- Reverabation
- Accent
- Dialect
- Disfluency
- Multi-lingual

Code-Switching Sentence Generation by Bert and Generative Adversarial Networks -China Mobile
Large-Scale Multilingual Speech Recognition with a Streaming End-to-End Model - Google
End-to-End ASR: From Supervised to Semi Supervised Learning with Modern Architecture - Facebook
Two-Pass End-to-End Speech Recognition - Google
RWTH ASR Systems for LibriSpeech: Hybrid vs Attention RWTH Aachen University
Multi-Task Self-Supervised Learning for Robust Speech Recognition - MILA
Two-stage Training for Chinese Dialect Recognition - Shanghai Institute
Noisy BiLSTM-based Models for
Disfluency Detection - Alibaba
Recognition of Intentions of Users' Short Responses for Conversational News Delivery System - Chiba, Japan
Personalisation of End-to-end Speech Recognition for Named Entities - Google
Semi-supervised DNN training with word selection for ASR - Brno Univ
SOTA ASR using Multi-Stream Self-Attention with Dilated 1d Convolutions - ASAPP
Seq2seq speech recognition with time-depth separable convolutions - Facebook
Source Guided Speaker Separation
Acoustic modelling based on mouth characteristics

Today: Transformer Based Approaches

Requirements:

- Multi-lingual
- Multi-Task
- Emotion detection
- Translation
- Include music, and other non-speech elements (coughing, giggling)

Personalised:

- Voice Cloning (based on few examples)

Data creation and selection - from large unlabelled data

- STT -> model -> TTS (models are good today)

Direction:

- One model with Flexibility to address the requirement

Transformer based Approaches:

- Adapter training for Audio Token Learning
  - **Model:** Whisper (earlier arch)
- **Audio token** (code book) - New Architecture
  - **Models:** Moshi, CSM, Orpheus
- **Architecture**
  - Hierarchical Transformer
  - Deep Transformer
  - Hierarchy of Audio Tokens
  - Creating Hierarchy from input

Thank You.